(FP2) 3. Differentiation 专题作业1

NO. 1

It is given that

$$x = \sinh^{-1} t,\ y = \cos^{-1} t,$$

where $-1 < t < 1$.

a. By differentiating $\cos y$ with respect to t, show that $\frac{dy}{dt} = -\frac{1}{\sqrt{1-t^2}}$. [4]

b. Find $\frac{d^2y}{dx^2}$ in terms of t, simplifying your answer. [5]

A level FURTHER MATHEMATICS 2020/Winter/Paper2_1/Q5

Mark scheme

Question	Answer	Marks	Guidance
a	$-\sin y\frac{\mathrm{d}y}{\mathrm{d}t}=1$	**M1** **A1**	Differentiates both sides with respect to t.
	$0 < y < \pi \Rightarrow \sin y > 0 \Rightarrow -\sqrt{1-\cos^2 y}\frac{\mathrm{d}y}{\mathrm{d}t}=1$	**M1**	Applies $\sin^2 y + \cos^2 y = 1$.
	$\frac{\mathrm{d}y}{\mathrm{d}t}=-\frac{1}{\sqrt{1-t^2}}$	**A1**	AG, justifies taking positive square root.
		4	
b	$\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{\sqrt{1+t^2}}$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}x}=-\sqrt{\frac{1+t^2}{1-t^2}}$	**B1**	Finds first derivative.
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(-\sqrt{\frac{1+t^2}{1-t^2}}\right)=-\frac{t\left(1-t^2\right)^{\frac{1}{2}}\left(1+t^2\right)^{-\frac{1}{2}}+t(1+t^2)^{\frac{1}{2}}\left(1-t^2\right)^{-\frac{1}{2}}}{1-t^2}$	**M1**	Differentiates $\frac{\mathrm{d}y}{\mathrm{d}x}$ with respect to t.
	$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=\frac{\mathrm{d}}{\mathrm{d}t}\left(-\sqrt{\frac{1+t^2}{1-t^2}}\right)\times\frac{\mathrm{d}t}{\mathrm{d}x}$	**M1**	Applies chain rule.
	$=-\frac{t\left(\left(1-t^2\right)^{\frac{1}{2}}+(1+t^2)\left(1-t^2\right)^{-\frac{1}{2}}\right)}{1-t^2}\quad\left(=-\frac{2t}{\left(1-t^2\right)^{\frac{3}{2}}}\right)$	**A1**	OE (simplified).
		5	

Explanation

a.

y = $\cos^{-1}t$ and $-1 < t < 1$, so $0 < y < \pi$. then sin y >0

$$\cos y = t$$

Differentiate respect to t:

$$-\sin y \frac{dy}{dt} = 1$$ **M1 A1**

$$\frac{dy}{dt} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-\cos^2 y}} = -\frac{1}{\sqrt{1-t^2}}$$ **M1A1**

b)

$$t = \sinh x$$

So $\cos y = \sinh x$

Differentiate respect to x twice:

$$-\sin y \frac{dy}{dx} = \cosh x \rightarrow \frac{dy}{dx} = -\frac{\cosh x}{\sin y}$$ **B1**

$$-\cos y \left(\frac{dy}{dx}\right)^2 - \sin y \frac{d^2y}{dx^2} = \sinh x$$ **B1**

Since ,$\sin y = \sqrt{1-\cos^2 y} = \sqrt{l-t^2}$,$\cosh^2 x = \sinh^2 x + 1 = t^2 + 1$ **M1**

$$\frac{d^2y}{dx^2} = -\frac{\sinh x}{\sin y} - \cos y \cdot \frac{\cosh^2 x}{\sin^2 y} \cdot \frac{1}{\sin y} = -\frac{t}{\sqrt{1-t^2}} - t \cdot \frac{1+t^2}{\sqrt{1-t^2}(1-t^2)}$$ **M1**

$$= -\frac{t(1-t^2)+t(1+t^2)}{(1-t^2)^{\frac{3}{2}}} = -\frac{2t}{(1-t^2)^{\frac{3}{2}}}$$ **A1**

Comment

a. Most candidates differentiated $\cos^{-1} t$ correctly using implicit differentiation.

b. The majority of candidates found the first derivative correctly using parametric differentiation. The attempts to find the second derivative varied in length, with stronger candidates showing the required level of algebraic fluency and remembering to divide by $\frac{dx}{dt}$ after differentiating with respect to t.

NO. 2

The curve C has parametric equations

$$x = \frac{1}{2}t^2 - \ln t,\ \ y = 2t + 1,\ \ \text{for } \frac{1}{2} \leq t \leq 2.$$

a. Find the exact length of C. [5]

b. Find $\frac{d^2y}{dx^2}$ in terms of t, simplifying your answer. [4]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_3/Q5

Mark scheme

Question	Answer	Marks
a	$\dot{x} = t - t^{-1},\ \dot{y} = 2$	**B1**
	$\dot{x}^2 + \dot{y}^2 = t^2 - 2 + t^{-2} + 4 = (t+t^{-1})^2$	**M1 A1**
	$\int_{\frac{1}{2}}^{2} t + t^{-1}\,\mathrm{d}t = \left[\frac{1}{2}t^2 + \ln t\right]_{\frac{1}{2}}^{2} = \frac{15}{8} + 2\ln 2$	**M1 A1**
		5
b	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dot{y}}{\dot{x}} = \frac{2}{t - t^{-1}}$	**B1**
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{2}{t-t^{-1}}\right) = \frac{-2\left(1+t^{-2}\right)}{\left(t-t^{-1}\right)^2}$	**B1**
	$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{2}{t-t^{-1}}\right) \times \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{-2\left(1+t^{-2}\right)}{\left(t-t^{-1}\right)^3}$	**M1 A1**
		4

Explanation

a.

Arc length is given by $\int ds = \int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$

$\frac{dx}{dt} = t - \frac{1}{t},\ \frac{dy}{dt} = 2$ **B1**

$$s = \int_{\frac{1}{2}}^{2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_{\frac{1}{2}}^{2} \sqrt{\left(t - \frac{1}{t}\right)^2 + (2)^2}\, dt$$

$$= \int_{\frac{1}{2}}^{2} \sqrt{t^2 - 2 + \frac{1}{t^2} + 4}\, dt == \int_{\frac{1}{2}}^{2} \sqrt{\left(t + \frac{1}{t}\right)^2}\, dt = \int_{\frac{1}{2}}^{2} \left(t + \frac{1}{t}\right) dt$$ **M1 A1**

$$= \left[\frac{1}{2}t^2 + \ln t\right]_{\frac{1}{2}}^{2} = \frac{1}{2}\left(4 - \frac{1}{4}\right) + \ln 2 - \ln\frac{1}{2} = \frac{15}{8} + 2\ln 2$$ **M1 A1**

b)

Use dy/dx = dy/dt ÷ dx/dt

$\frac{dy}{dx} = \frac{2}{t - \frac{1}{t}} = \frac{2t}{t^2 - 1}$ **B1**

$\frac{dt}{dx} = \frac{1}{t - \frac{1}{t}} = \frac{t}{t^2 - 1}$(reciprocal of dx/dt) **B1**

$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{dt}{dx} = \frac{d}{dt}\left(\frac{2t}{t^2 - 1}\right) \cdot \frac{t}{t^2 - 1} = \frac{2(t^2 - 1) - 2t \cdot 2t}{(t^2 - 1)^2} \cdot \frac{t}{t^2 - 1} =$ **M1 A1**

$-\frac{2t^3 + 2t}{(t^2 - 1)^3}$

NO. 3

The variables x and y are such that $y = -1$ when $x = 0$ and

$$\left(x + \frac{dy}{dx}\right)^3 = y^2 + x.$$

i. Find the value of $\frac{dy}{dx}$ when $x = 0$. [1]

ii. Find also the value of $\frac{d^2y}{dx^2}$ when $x = 0$. [4]

A level FURTHER MATHEMATICS 2018/Summer/Paper1_3/Q1

Mark scheme

Question	Answer	Marks	Guidance
1.i	$\left(0+\frac{dy}{dx}\right)^3=(-1)^2+0\Rightarrow\frac{dy}{dx}=1$	**B1**	
ii	$3\left(x+\frac{dy}{dx}\right)^2\left(1+\frac{d^2y}{dx^2}\right)$	**M1 A1**	Method mark for good attempt at implicit differentiation of LHS.
	$=2y\frac{dy}{dx}+1$	**B1**	Note: may expand bracket before differentiation but M1 is still for implicit differentiation
	$\Rightarrow 3\left(1+\frac{d^2y}{dx^2}\right)=-2+1\Rightarrow\frac{d^2y}{dx^2}=-\frac{4}{3}$	**A1**	
		5	

Explanation

1.i

when x=0, y=-1

$\left(x+\frac{dy}{dx}\right)^3=y^2+x$

$\rightarrow\left(0+\frac{dy}{dx}\right)^3=(-1)^2+0$

$\left(\frac{dy}{dx}\right)^3=1$

$\frac{dy}{dx}=1$ **B1**

ii)

Differentiate the given equation:

$$3\left(x+\frac{dy}{dx}\right)^2\cdot\left(1+\frac{d^2y}{dx^2}\right)=2y\frac{dy}{dx}+1$$ **M1 A1 B1**

when x=0, y=-1, $\frac{dy}{dx}=1$

$$3\left(0+(-1)\right)^2\cdot\left(1+\frac{d^2y}{dx^2}\right)=2\cdot(-1)\cdot(1)+1$$

$$3\left(1+\frac{d^2y}{dx^2}\right)=-1$$

$$1+\frac{d^2y}{dx^2}=-\frac{1}{3}$$

$$\frac{d^2y}{dx^2}=-\frac{4}{3}$$ **A1**

Comment

i. The first part of this question did not pose any difficulties.
ii. Candidates who used the chain rule were usually successful. Those who chose to multiply out the expression were at risk of errors, particularly if they multiplied the brackets out rather than using a binomial expansion. A small number of candidates took the cube root of both sides of the equation before differentiating, and were usually successful too.

Answers: **(i)** $\frac{dy}{dx}=1$ **(ii)** $\frac{d^2y}{dx^2}=-\frac{4}{3}$

NO. 4

A curve C has equation $\cos y = x$, for $-\pi < x < \pi$.

i. Use implicit differentiation to show that $\frac{d^2y}{dx^2} = -\cot y\left(\frac{dy}{dx}\right)^2$. [4]

ii. Hence find the exact value of $\frac{d^2y}{dx^2}$ at the point $\left(\frac{1}{2}, \frac{1}{3}\pi\right)$ on C. [2]

A level FURTHER MATHEMATICS 2019/Summer/Paper1_1/Q1

Mark scheme

Question	Answer	Marks	Guidance
1.i	$-\sin y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -(\sin y)^{-1}$	**M1 A1**	Differentiates implicitly once.
	$\frac{d^2y}{d^2x} = (\sin y)^{-2}\cos y\left(\frac{dy}{dx}\right) = -\cot y\left(\frac{dy}{dx}\right)^2$	**M1 A1**	Differentiates again, AG.
		4	
ii	$\frac{dy}{dx} = -\left(\sin\frac{\pi}{3}\right)^{-1} = -\frac{2}{\sqrt{3}}$	**B1**	
	$\frac{d^2y}{d^2x} = -\frac{1}{\sqrt{3}}\left(-\frac{2}{\sqrt{3}}\right)^2 = -\frac{4}{3\sqrt{3}} = -\frac{4}{9}\sqrt{3}$	**B1**	AEF, must be exact.
		2	

Explanation

1.i

Use implicit differentiation

$$\cos y = x$$

$$-\sin y \cdot \frac{dy}{dx} = 1 \quad \textbf{M1 A1}$$

Differentiate again:

$$-\cos y \cdot \frac{dy}{dx} \cdot \frac{dy}{dx} - \sin y \frac{d^2y}{dx^2} = 0$$

$$\frac{d^2y}{dx^2} = -\cot y \left(\frac{dy}{dx}\right)^2 \quad \textbf{M1 A1}$$

ii)

use results in i)

$$\frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sin\frac{\pi}{3}} = -\frac{2}{\sqrt{3}} \quad \textbf{B1}$$

$$\frac{d^2y}{dx^2} = -\cot y \left(\frac{dy}{dx}\right)^2 = -\frac{1}{\tan\frac{\pi}{3}}\left(-\frac{2}{\sqrt{3}}\right)^2 = -\frac{1}{\sqrt{3}} \cdot \frac{4}{3} = -\frac{4\sqrt{3}}{9} \quad \textbf{B1}$$

Comment

i. This was very well done and good knowledge of implicit differentiation was seen.

ii. Many candidates were accurate in evaluating the second derivative, though a few made errors with signs.

NO. 5

A curve C has equation $x^3 - 3xy + y^2 = 4$. Find the value of $\frac{d^2y}{dx^2}$ at the point (0, 2) of C. [7]

A level FURTHER MATHEMATICS 2017/Summer/Paper1_3/Q4

Mark scheme

Question	Answer	Marks	Guidance
4	$3x^2 - \left(3y + 3x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$	**B1B1**	SOI
	At (0,2): $-6 + 4\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{3}{2}$	**B1**	
	$6x - 3\frac{dy}{dx} - \left(3\frac{dy}{dx} + 3x\frac{d^2y}{dx^2}\right)$	**M1A1**	
	$+2\left(\frac{dy}{dx}\right)^2 + 2y\frac{d^2y}{dx^2} = 0$	**A1**	
	At (0,2): $-\frac{9}{2} - \frac{9}{2} + \frac{9}{2} + 4\frac{d^2y}{dx^2} = 0 \Rightarrow \frac{d^2y}{dx^2} = \frac{9}{8}$	**A1**	
	Total:	**7**	

Explanation

4.

Differentiate the equation respect to x:

$$3x^2 - \left(3y + 3x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$$ **B1B1**

At (0,2): $-6 + 4\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{3}{2}$ **B1**

Differentiate respect to x again:

$$6x - 3\frac{dy}{dx} - 3\frac{dy}{dx} - 3x\frac{d^2y}{dx^2} + \left(2\frac{dy}{dx}\cdot\frac{dy}{dx} + 2y\frac{d^2y}{dx^2}\right) = 0$$

M1A1 A1

At (0,2): $-\frac{9}{2} - \frac{9}{2} + \frac{9}{2} + 4\frac{d^2y}{dx^2} = 0 \Rightarrow \frac{d^2y}{dx^2} = \frac{9}{8}$ **A1**

Comment

This question was very well done by most candidates. They showed a good grasp of implicit differentiation and where they chose to rearrange their expression for $\frac{dy}{dx}$ they also used the quotient rule accurately. Most candidates were accurate in their handling of the necessary algebra.

Answer: $\frac{d^2y}{dx^2} = \frac{9}{8}$

NO. 6

A curve C has equation $\tan y = x$, for $x > 0$.

i. Use implicit differentiation to show that $\frac{d^2y}{dx^2} = -2x\left(\frac{dy}{dx}\right)^2$. [3]

ii. Hence find the value of $\frac{d^2y}{dx^2}$ at the point $(1, \frac{1}{4}\pi)$ on C. [2]

A level FURTHER MATHEMATICS 2017/Summer/Paper1_1/Q3

Mark scheme

Question	Answer	Marks	Guidance
3.i	$(1 + x^2)\frac{dy}{dx} = 1$ **or** $\sec^2 y\frac{dy}{dx} = 1 \Rightarrow (1 + x^2)\frac{dy}{dx} = 1$ **or** $\frac{dy}{dx} = \cos^2 y$	**M1**	Using implicit differentiation
	$\Rightarrow 2x\frac{dy}{dx} + (1 + x^2)\frac{d^2y}{dx^2} = 0 \Rightarrow \frac{d^2y}{dx^2} = -2x(\frac{dy}{dx})^2$ (AG)	**M1** **A1**	M1 for good attempt at product rule
	Alt method: $\frac{dy}{dx} = \cos^2 y \Rightarrow \frac{d^2y}{dx^2} = \cos y(-\sin y)\frac{dy}{dx} = -2x(\frac{dy}{dx})^2$	**M1** **A1**	M1 for good attempt at implicit differentiation
	Total:	**3**	
ii	$y'(1) = \cos^2\left(\frac{\pi}{4}\right) \Rightarrow y'(1) = \frac{1}{2}$	**B1**	
	$\Rightarrow y''(1) = -\frac{1}{2}$	**B1 FT**	
	Total:	**2**	

Explanation

3.i

differentiate respect to x: $tan\, y = x$

$\sec^2 y \frac{dy}{dx} = 1$ ($sec^2 y = \tan^2 y + 1 = x^2 + 1$) **M1**

$$(x^2+1)\frac{dy}{dx} = 1$$

$$2x \cdot \frac{dy}{dx} + (x^2+1)\frac{d^2y}{dx^2} = 0$$

$$\frac{d^2y}{dx^2} = -\frac{2x}{x^2+1}\frac{dy}{dx} = -2x\left(\frac{dy}{dx}\right)\frac{dy}{dx} = -2x\left(\frac{dy}{dx}\right)^2$$ **M1 A1**

$(\frac{dy}{dx} = \frac{1}{x^2+1})$

3.ii

$$\frac{d^2y}{dx^2} = -2x\left(\frac{dy}{dx}\right)^2 = -\frac{2x}{(x^2+1)^2}$$ **B1**

When x=1

$$\frac{d^2y}{dx^2} = -\frac{2}{(1^2+1)^2} = -\frac{1}{2}$$ **B1**

Comment

In **(i)**, most students used implicit differentiation as required and deduced the given relationship, rearranging expressions if necessary. The standard of differentiation was generally very good. Better candidates were careful in **(ii)** to find $\frac{dy}{dx}$ and then square it, as required by the given formula.

Answer: **(ii)** $\frac{dy}{dx} = \cos^2\left(\frac{\pi}{4}\right) = \frac{1}{2}, \frac{d^2y}{dx^2} = -\frac{1}{2}$

NO. 7

i. A curve C has equation $x^2 + 4xy - y^2 + 20 = 0$. Show that, at stationary points on C, $x = -2y$. [3]

ii. Find the coordinates of the stationary points on C, and determine their nature by considering the value of $\frac{d^2y}{dx^2}$ at the stationary points. [8]

A level FURTHER MATHEMATICS 2016/Winter/Paper1_1/Q8

Mark scheme

8.i	$2x + 4(xy' + y) - 2yy' = 0$ $(*)$	**M1A1**	**[3]**
	$y' = 0 \Rightarrow 2x + 4y = 0 \Rightarrow x = -2y$ (AG)	**A1**	
ii	At stationary points $4y^2 - 8y^2 - y^2 + 20 = 0 \Rightarrow 5y^2 = 20 \Rightarrow y = \pm 2$	**M1A1**	**[8]**
	Coordinates of stationary points are $(4, -2)$ and $(-4, 2)$	**A1**	
	From $(*)$: $x + 2(xy' + y) - yy' = 0$ Differentiating: $1 + 2(xy'' + y' + y') - (yy'' + [y']^2) = 0$ (or by quotient rule)	**M1A1**	
	At $(4, -2)$ with $y' = 0$: $1 + 8y'' + 2y'' = 0 \Rightarrow y'' = -\frac{1}{10} \Rightarrow$ maximum.	**M1A1**	
	At $(-4, 2)$ with $y' = 0$: $1 - 8y'' - 2y'' = 0 \Rightarrow y'' = \frac{1}{10} \Rightarrow$ minimum.	**A1**	

Explanation

8.i

differentiate respect to x:

$$2x + \left(4y + 4x\frac{dy}{dx}\right) - 2y\frac{dy}{dx} = 0$$ **M1A1**

Stationary points: $\frac{dy}{dx} = 0$

$$2x + 4y = 0$$

x=-2y

A1

ii.

Substitute x=-2y into $x^2 + 4xy - y^2 + 20 = 0$:

$$4y^2 - 8y^2 - y^2 + 20 = 0$$

$$5y^2 = 20$$

$$y = \pm 2, x = \mp 4$$ **M1A1**

Stationary points: (-4,2), (4,-2) **A1**

Differentiate respect to x again:

$$2 + 4\frac{dy}{dx} + \left(4\frac{dy}{dx} + 4x\frac{d^2y}{dx^2}\right) - \left(2\frac{dy}{dx}\cdot\frac{dy}{dx} + 2y\frac{d^2y}{dx^2}\right) = 0$$ **M1A1**

For the point (-4,2)

$$2 + 4(-4)\frac{d^2y}{dx^2} - 2(2)\frac{d^2y}{dx^2} = 0$$

$$\frac{d^2y}{dx^2} = \frac{1}{10} > 0$$

So (-4,2) is a local minimum **M1A1**

For the point (4,-2)

$$2 + 4(4)\frac{d^2y}{dx^2} - 2(-2)\frac{d^2y}{dx^2} = 0$$

$\frac{d^2y}{dx^2} = -\frac{1}{10} < 0$

So (-4,2) is a local maximum **A1**

Comment

Many candidates completed this question effectively, applying implicit differentiation accurately to find both first and second derivatives. Several found the coordinates of the stationary points correctly, but switched the x and y coordinates round when determining whether the points were maximum or minimum.

Answer: Maximum at (4, –2) and minimum at (–4, 2)

NO. 8

It is given that

$$x = 1 + \frac{1}{t} \quad \text{and} \quad y = \cos^{-1} t \quad \text{for } 0 < t < 1.$$

a. Show that $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{t^2}{\sqrt{1-t^2}}$. [2]

b. Show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -t^a\left(1-t^2\right)^b\left(2-t^2\right)$, where a and b are constants to be determined. [4]

A level FURTHER MATHEMATICS 2024/Winter/Paper2_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
a	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t} \times \frac{\mathrm{d}t}{\mathrm{d}x} = -\frac{1}{\sqrt{1-t^2}} \times \left(-t^2\right) = \frac{t^2}{\sqrt{1-t^2}}$	M1A1	Uses chain rule, AG.
		2	
b	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) = \frac{2t\sqrt{1-t^2} + t^3\left(1-t^2\right)^{-\frac{1}{2}}}{1-t^2}$	M1A1	Applies Quotient rule.
	$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{2t\sqrt{1-t^2} + t^3\left(1-t^2\right)^{-\frac{1}{2}}}{1-t^2}\left(-t^2\right) = -2t^3\left(1-t^2\right)^{-\frac{1}{2}} - t^5\left(1-t^2\right)^{-\frac{3}{2}}$ $= -t^3\left(1-t^2\right)^{-\frac{3}{2}}\left(2\left(1-t^2\right)+t^2\right) = -t^3\left(1-t^2\right)^{-\frac{3}{2}}\left(2-t^2\right)$	M1A1	Uses chain rule.
		4	

Comment

a. Almost all candidates applied the chain rule to find the first derivative.
b. Most candidates differentiated the first derivative with respect to t and applied the chain rule again to obtain the correct result.

NO. 9

It is given that

$$x = \sin^{-1}t \quad \text{and} \quad y = t\cos^{-1}t, \quad \text{for } 0 \leqslant t < 1.$$

a. Show that $\frac{dy}{dx} = -t + \sqrt{1-t^2}\cos^{-1}t$. [3]

b. Find $\frac{d^2y}{dx^2}$ in terms of t. [4]

A level FURTHER MATHEMATICS 2024/Summer/Paper2_1/Q3

Mark scheme

Question	Answer	Marks	Guidance
a	$\frac{dy}{dt} = \cos^{-1}t - \frac{t}{\sqrt{1-t^2}}$, $\frac{dx}{dt} = \frac{1}{\sqrt{1-t^2}}$	B1	
	$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = -t + \sqrt{1-t^2}\cos^{-1}t$	M1 A1	Uses chain rule, AG.
		3	
b	$\frac{d}{dt}\left(\frac{dy}{dx}\right) = -1 - \frac{\sqrt{1-t^2}}{\sqrt{1-t^2}} - \frac{t\cos^{-1}t}{\sqrt{1-t^2}} = -2 - \frac{t\cos^{-1}t}{\sqrt{1-t^2}}$	M1 A1	Applies product rule.
	$\frac{d^2y}{dx^2} = -2\sqrt{1-t^2} - t\cos^{-1}t$	M1 A1	Uses chain rule.
		4	

Comment

a. The majority of candidates found the first derivative correctly using parametric differentiation, although sign errors occasionally occurred when deriving the given answer.
b. The attempts to find the second derivative varied in length, with strong candidates showing the required level of algebraic fluency and remembering to divide by $\frac{dx}{dt}$ after differentiating with respect to t.

NO. 10

It is given that

$$x = 1 + \frac{1}{t} \qquad \text{and} \qquad y = te^t.$$

a. Show that $\frac{dy}{dx} = -e^t(t^3 + t^2)$. [3]

b. Find $\frac{d^2y}{dx^2}$ in terms of t. [4]

A level FURTHER MATHEMATICS 2023/Winter/Paper2_2/Q2

Mark scheme

Question	Answer	Marks	Guidance
a	$\frac{dy}{dt} = e^t(t+1),\ \ \frac{dx}{dt} = -t^{-2}$	B1	
	$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = -t^2 e^t(t+1) = -e^t(t^3+t^2)$	M1 A1	Uses chain rule, AG.
		3	
b	$\frac{d}{dt}\left(\frac{dy}{dx}\right) = -e^t(3t^2+2t) - e^t(t^3+t^2) = -e^t(t^3+4t^2+2t)$	M1 A1	Applies product rule.
	$\frac{d^2y}{dx^2} = t^3 e^t(t^2+4t+2)$	M1 A1	Uses chain rule, multiplies by *their* $\frac{dt}{dx}$ for M1.
	Alternative method for question 2(b)		
	$\frac{d^2y}{dx^2} = -e^t\left(3t^2\frac{dt}{dx} + 2t\frac{dt}{dx}\right) - e^t(t^3+t^2)\frac{dt}{dx}$	(M1 A1)	Differentiates $-e^t(t^3+t^2)$ t implicitly.
	$\frac{d^2y}{dx^2} = e^t(-4t^2-2t-t^3)\frac{dt}{dx} = t^3e^t(t^2+4t+2)$	(M1 A1)	Substitutes $\frac{dt}{dx} = -t^2$.
		4	

Comment

a. The majority of candidates found the first derivative correctly using parametric differentiation.

b. Most candidates were able to find $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)$ using the product rule, but some omitted to multiply by $-t^2$ to obtain the second derivative with respect to x as required.